

1 • INCLUSION BODIES — THE MASTER TABLE ★★★

Named body	Site	Disease
Negri body	Intra-CYTOPLASMIC	RABIES — hippocampal pyramidal cells & cerebellar Purkinje cells
Cowdry type A	Intra-NUCLEAR	Herpesviruses (IBR, EHV-1) and adenovirus (ICH)
Bollinger body	Intra-cytoplasmic	FOWL POX (contains Borrel bodies = elementary bodies)
Guarnieri body	Intra-cytoplasmic	Vaccinia / cowpox / buffalopox
Borrel body	Intra-cytoplasmic	Elementary bodies of pox viruses
CDV inclusions	BOTH nuclear AND cytoplasmic	Canine distemper — epithelium, astrocytes, neurons, bladder
Marek's — no inclusion	—	Lymphoproliferative, not inclusion-forming

RULE OF THUMB DNA viruses replicate in the nucleus → **INTRANUCLEAR** inclusions (herpes, adeno, parvo). **Exception: POXVIRUS** is a DNA virus that replicates in the **CYTOPLASM** → **intracytoplasmic**. RNA viruses → **intracytoplasmic** (rabies, distemper) — but paramyxoviruses (CDV, rinderpest) can give **both**.

2 • FOOT AND MOUTH DISEASE ★★★

- **Aphthovirus**, family **Picornaviridae**; ssRNA. **7 serotypes** — **O, A, C, SAT-1, SAT-2, SAT-3, Asia-1**. In India the circulating types are **O, A and Asia-1**. **No cross-protection between serotypes** — this is why vaccination fails when the type changes
- Cloven-footed animals. **Very high morbidity (up to 100%), low mortality in adults** — but **high mortality in young stock**
- **Gross: vesicles (aphthae)** → ruptured, leaving red raw erosions, on the **tongue, dental pad, gums, muzzle, interdigital space, coronary band and teats**; profuse ropy salivation, lip-smacking, lameness
- **"TIGER HEART"** ★ in calves — pale yellow-grey streaks of myocardial degeneration and necrosis alternating with red muscle; this is what kills the young
- **Micro: intra-epithelial vesicle** from **hydropic and ballooning degeneration of the stratum spinosum**
- **NOTIFIABLE**. Sample: **vesicular fluid and epithelial flap in 50% glycerol-phosphate buffer, pH 7.2–7.6**, sent chilled

THE 4 VESICULAR DISEASES LOOK IDENTICAL ★★ FMD — cattle, sheep, goat, pig; **NOT horses**. **Vesicular stomatitis** — the only one that **DOES affect HORSES** (and is zoonotic). **Vesicular exanthema** and **swine vesicular disease** — **pigs only**. They can be separated only in the laboratory. **The horse is the field clue**.

3 • RINDERPEST • PPR • MCF

Disease	Agent	Signature lesion
RINDERPEST (cattle plague)	Morbillivirus , Paramyxoviridae	"ZEBRA STRIPING" — linear congestion and haemorrhage on the longitudinal folds of the colon and rectum . Necrotic erosive stomatitis, "shooting" diarrhoea, Peyer's patch necrosis , syncytial giant cells with both nuclear and cytoplasmic inclusions . Globally eradicated (declared 2011); India declared free in 2006
PPR (goat plague)	Morbillivirus	Stomatitis–pneumoenteritis of goats more than sheep . Erosive stomatitis, mucopurulent oculo-nasal discharge, zebra striping of colon, bronchointerstitial pneumonia . Looks like rinderpest but in small ruminants
MALIGNANT CATARRHAL FEVER	Ovine herpesvirus-2 (sheep-associated, the Indian form) or alcelaphine herpesvirus-1 (wildebeest-associated)	BILATERAL CORNEAL OPACITY spreading from the LIMBUS INWARD — the classic "head and eye" form. Generalised lymphadenopathy , erosive stomatitis and rhinitis. THE microscopic key: widespread lymphocytic NECROTISING VASCULITIS/ARTERITIS in every organ . Sporadic, almost always fatal , cattle are dead-end hosts

4 • BLUETONGUE • IBR • BVD-MD

Disease	Agent & key facts
BLUETONGUE	Orbivirus , Reoviridae ; transmitted by Culicoides midges, therefore seasonal (rainy season) and non-contagious . SHEEP are worst affected. Facial and submandibular oedema, cyanotic swollen protruding tongue , erosions of the buccal mucosa, coronitis with a red-purple band at the coronary band and later hoof separation, torticollis. ★ Haemorrhage at the BASE OF THE PULMONARY ARTERY is the characteristic post-mortem finding . Skeletal muscle necrosis
IBR / IPV	Bovine herpesvirus-1 . Respiratory form ("red nose") — necrotic grey-white plaques on the nasal mucosa, laryngotracheitis with a fibrinonecrotic cast. Genital form — infectious pustular vulvovaginitis / balanoposthitis. Abortion — usually in the last trimester, autolysed fetus with pinpoint necrotic foci in the liver . Also conjunctivitis and calf encephalitis. Cowdry type A intranuclear inclusions early. Establishes LATENCY in the trigeminal ganglion and recrudesces with stress
BVD / MUCOSAL DISEASE ★	Pestivirus , Flaviviridae . Two biotypes: non-cytopathic (ncp) and cytopathic (cp) . The PI story — learn this: infection of the dam with the ncp biotype at about 40–120 days of gestation , before the fetal immune system matures, produces a calf that is IMMUNOTOLERANT and PERSISTENTLY INFECTED (PI) . That PI animal sheds virus for life and develops fatal MUCOSAL DISEASE if it is later superinfected by the CYTOPATHIC biotype . Lesions: linear erosions/ulcers of the oesophagus (highly suggestive), erosions from mouth to colon, Peyer's patch necrosis , interdigital erosions, lymphoid depletion. Infection at about 100–170 days instead causes CEREBELLAR HYPOPLASIA and ocular defects

5 · EQUINE VIRAL DISEASES

Disease	Agent & lesion
Equine infectious anaemia ("swamp fever")	Lentivirus, Retroviridae ; biting flies and contaminated needles. Recurrent fever, progressive anaemia, ventral oedema, emaciation; LIFELONG CARRIER . Pathogenesis is immune-mediated : repeated antigenic drift → immune complexes → Type III glomerulonephritis and Coombs-positive haemolysis. Micro: SIDEROLEUKOCYTES (haemosiderin-laden macrophages) in liver, spleen and bone marrow; lymphoid proliferation around vessels. Diagnosis: Coggins test (AGID) and cELISA
Equine influenza	Influenza A , subtypes H7N7 and H3N8 . Sudden dry, harsh, hacking cough , high fever, rapid spread; damage to the tracheal cilia predisposes to secondary bacterial pneumonia
Equine viral arteritis	Arterivirus . Panvascular disease — necrotising arteritis of small muscular arteries → oedema of limbs, ventral abdomen and periorbital tissue ("pink eye"), abortion at any stage. Stallions become long-term carriers and shed in semen
Equine rhinopneumonitis	EHV-1 and EHV-4 . EHV-4 → mainly upper respiratory disease in foals. EHV-1 → abortion storms in the last trimester ; the aborted fetus is FRESH (not autolysed) , icteric, with pleural fluid and pinpoint white necrotic foci in the liver with Cowdry type A intranuclear inclusions ; also myeloencephalopathy from vasculitis and thrombosis
Equine encephalomyelitis	Alphavirus (Togaviridae) — EEE, WEE, VEE ; mosquito-borne, birds are the reservoir, zoonotic . Non-suppurative polioencephalomyelitis — neuronal necrosis, neuronophagia, perivascular cuffing, gliosis, chiefly in grey matter

6 · SWINE VIRAL DISEASES

CLASSICAL SWINE FEVER — THE FOUR-LESION SET ★★★ Pestivirus (same genus as BVD). Highly contagious, high mortality.

- ① **"BUTTON ULCERS" (botryomycotic buttons)** — raised, laminated, concentric necrotic ulcers of the **caecum, colon and ileocaecal valve**
- ② **MARGINAL SPLENIC INFARCTS** — dark wedge-shaped infarcts on the splenic border (near pathognomonic)
- ③ **"TURKEY-EGG KIDNEY"** — petechiae scattered over a pale renal cortex; petechiae also in bladder, larynx and skin
- ④ **HAEMORRHAGIC ("MARBLED") LYMPH NODES** — haemorrhage at the periphery of the node
Plus **non-suppurative encephalitis with perivascular cuffing**.
Congenital infection → **cerebellar hypoplasia and congenital tremor**

Swine influenza — Influenza A, usually **H1N1**; explosive outbreak, high morbidity, low mortality; **cranioventral purple-red consolidation**, necrotising bronchiolitis; **zoonotic potential**.

7 · CANINE & FELINE VIRAL DISEASES ★★

Disease	Agent & key lesion
RABIES	Lyssavirus, Rhabdoviridae, bullet-shaped . Virus in saliva → centripetal spread along peripheral nerves to the CNS, then centrifugal spread to the salivary glands (that is why the bite is infective). Forms: furious, dumb/paralytic . Gross lesions are minimal or absent — foreign bodies in a dog's stomach are suggestive. Micro: non-suppurative polioencephalitis + NEGRI BODIES (intracytoplasmic, in hippocampal pyramidal neurons of carnivores and cerebellar Purkinje cells of herbivores); Babes' nodules = glial nodules. Diagnosis: FAT on brain impression smear is the GOLD STANDARD ; also Seller's stain for Negri bodies, mouse inoculation, RT-PCR. Zoonosis — 100% fatal once signs appear
CANINE DISTEMPER	Morbillivirus . Pantropic — respiratory, GI, skin, CNS. Biphasic fever , oculo-nasal discharge, bronchointerstitial pneumonia , "HARD PAD" (nasodigital hyperkeratosis) , enamel hypoplasia ("distemper teeth") if infected before teeth erupt, demyelinating leukoencephalitis , "chewing-gum" fits. Inclusions are BOTH intranuclear and intracytoplasmic , best seen in bladder, gastric and bronchial epithelium and astrocytes . Marked lymphoid depletion → secondary infection
INFECTIOUS CANINE HEPATITIS	Canine adenovirus-1 . Centrilobular hepatic necrosis with Cowdry type A intranuclear inclusions in hepatocytes and endothelium ; OEDEMA OF THE GALL-BLADDER WALL is characteristic ; fibrin strands on the liver capsule; petechiae; "BLUE EYE" = corneal oedema and anterior uveitis, a Type III immune-complex reaction during recovery or after live vaccine
CANINE PARVOVIRUS (CPV-2)	Attacks rapidly dividing cells . Intestinal CRYPT NECROSIS (not the villus tip) → villus collapse, haemorrhagic enteritis with foul bloody diarrhoea and vomiting; lymphoid depletion and marked leucopenia ; myocarditis in pups infected under about 8 weeks. Intranuclear inclusions
FELINE PANLEUKOPENIA	Feline parvovirus. Severe panleukopenia , crypt necrosis and enteritis, and CEREBELLAR HYPOPLASIA in kittens infected in utero or just after birth

8 • OVINE RETROVIRAL LUNG DISEASE • ROTA & CORONA • PRIONS

Disease	Key point
MAEDI (– visna)	Ovine lentivirus. Maedi = chronic progressive INTERSTITIAL pneumonia — lungs heavy, grey-blue, rubbery, DO NOT COLLAPSE , with rib impressions and lymphoid follicle formation. Visna = demyelinating meningoencephalitis . Long incubation, no fever, no cure
JAAGSIEKTE ★	Ovine pulmonary adenocarcinoma , a betaretrovirus — a genuinely CONTAGIOUS LUNG TUMOUR . Firm grey-white nodules, later confluent; abundant fluid. "WHEELBARROW TEST" — lifting the hindquarters makes copious frothy fluid pour from the nostrils
Rotavirus	Neonatal diarrhoea. Infects MATURE enterocytes at the VILLUS TIP → villous atrophy and blunting → malabsorptive, osmotic diarrhoea . Crypts are spared, so recovery is possible
Coronavirus	TGE of pigs — very severe villous atrophy, near-100% mortality under 2 weeks. Bovine coronavirus — calf diarrhoea and winter dysentery in adults. Also feline coronavirus → FIP with pyogranulomatous vasculitis
PRION DISEASES ★	The agent is PrP^{Sc} — an abnormally folded PROTEIN with NO nucleic acid . Therefore: no inflammation, no immune response, no antibody, no fever , and extreme resistance to formalin, heat, UV and ordinary disinfectants . Very long incubation, invariably fatal. Common lesion: STATUS SPONGIOSUS — vacuolation of neurons and neuropil in the grey matter, gliosis, neuronal loss, amyloid plaques . SCRAPIE (sheep, goats) — intense pruritus , wool loss, ataxia. BSE (cattle, "mad cow") — spread by feeding meat-and-bone meal ; apprehension, hyperaesthesia, ataxia; linked to variant CJD in people. FSE in cats; chronic wasting disease in deer. Diagnosis: histopathology of the brain stem (obex) + IMMUNOHISTOCHEMISTRY or Western blot for PrP

9 • POX VIRUS DISEASES

Macul e	→	Papule	→	Vesicle	→	Pustul e	→	Crust / scab	→	Scar
• Poxviruses are large DNA viruses that replicate in the CYTOPLASM, hence intracytoplasmic inclusions; micro shows ballooning degeneration of keratinocytes and epidermal hyperplasia • SHEEP POX — the most severe pox of domestic animals (Capripoxvirus); generalised skin nodules, and hard grey nodules in the LUNG are characteristic; "sheep-pox cells" with vacuolated nuclei • GOAT POX — similar, milder; LUMPY SKIN DISEASE (capripox, cattle) — multiple firm intradermal nodules over the whole body, "sit-fast" cores, lymphadenopathy, oedema of the limbs • Cowpox / BUFFALOPOX — teat and udder lesions, zoonotic, important in Indian dairies (milkers' nodules) • ORF (contagious ecthyma / contagious pustular dermatitis) — a PARAPOXVIRUS of sheep and goats; proliferative, scabby, wart-like lesions on the lips, commissures and muzzle of lambs; zoonotic; interferes with suckling • Swine pox — usually spread by the pig louse <i>Haematopinus suis</i>; • Fowl pox — cutaneous (comb, wattle) and diphtheritic forms, Bollinger bodies										

10 • TUBERCULOSIS & JOHNE'S DISEASE ★★★

	Tuberculosis	Johne's disease
Agent	<i>M. bovis</i> (cattle), <i>M. tuberculosis</i> , <i>M. avium</i>	<i>M. avium</i> subsp. <i>paratuberculosis</i>
Reaction	Granuloma (tubercle) with CASEOUS necrosis + epithelioid cells + Langhans giant cells + lymphocyte cuff + fibrous capsule; calcification is prominent in cattle	DIFFUSE granulomatous enteritis — NO caseation, NO calcification, NO ulceration; sheets of epithelioid macrophages stuffed with acid-fast bacilli in the lamina propria
Gross	Miliary nodules; "PEARL DISEASE" = grape-like nodules on pleura and peritoneum; Ghon (primary) complex = the primary lung lesion + its regional lymph node	Thickened, CORRUGATED / rugose ileum that will not flatten, "CORDING" of the subserosal lymphatics, enlarged oedematous mesenteric lymph nodes
Signs	Chronic wasting, cough	Chronic intractable diarrhoea with a normal appetite, progressive emaciation, bottle jaw; adults, long incubation
Stain / test	Ziehl–Neelsen ACID-FAST; tuberculin test = Type IV delayed hypersensitivity	ZN acid-fast in clumps; johnin test, faecal culture, ELISA, PCR

11 · ANTHRAX & THE CLOSTRIDIA ★★★

ANTHRAX *Bacillus anthracis* — Gram-positive, **capsulated, spore-forming** rod; spores survive in soil for decades. **Peracute death** in ruminants. **Dark tarry blood that fails to clot oozes from the nostrils, mouth, anus and vulva; rigor mortis is absent or incomplete; the SPLEEN is huge, soft and blackberry-jam-like.** Widespread haemorrhage and oedema.

NEVER open the carcass. Make a **peripheral blood smear** and stain with **polychrome methylene blue** → **McFADYEAN'S REACTION** (blue rods with a pink capsule). Dispose by **deep burial with quicklime or incineration**. Notifiable, zoonotic (malignant pustule, woolsorter's disease). In **pigs** it is usually a localised **pharyngeal/oedematous** form; in horses, colic and oedema.

Clostridial disease	Organism & the fact that identifies it
BLACK QUARTER (blackleg)	<i>Cl. chauvoei</i> . ENDOGENOUS — spores already in the muscle germinate after bruising, no wound needed . Young, well-nourished cattle 6 months–2 years . Heavy muscle becomes dark red-black, dry, spongy, CREPITANT with a rancid-butter odour ; hot painful swelling that becomes cold and crackly. Prevented by vaccination before the rains
Malignant oedema	<i>Cl. septicum</i> — EXOGENOUS , needs a WOUND ; more oedematous, less gassy than blackleg
BLACK DISEASE (infectious necrotic hepatitis)	<i>Cl. novyi</i> type B in sheep. Requires liver damage by migrating immature <i>Fasciola</i> to create the anaerobic focus. Large pale-yellow necrotic foci surrounded by a hyperaemic zone in the liver; after death intense subcutaneous venous congestion turns the skin dark — hence the name
ENTEROTOXAEMIA	<i>Cl. perfringens</i> types. Type D = "PULPY KIDNEY" of lambs — epsilon toxin after a sudden change to rich feed; GLYCOSURIA is a useful antemortem clue; kidneys soften rapidly after death; focal symmetrical encephalomalacia . Type B = lamb dysentery; Type C = "struck" and necrotic enteritis
BRAXY	<i>Cl. septicum</i> — ABOMASITIS of sheep that eat frosted/frozen grass ; haemorrhagic necrotic abomasal wall
BOTULISM	<i>Cl. botulinum</i> — an INTOXICATION : the preformed toxin is EATEN . Toxin blocks acetylcholine release at the neuromuscular junction → FLACCID paralysis , tongue paralysis, "limberneck" in birds. Type D in cattle follows PHOSPHORUS deficiency → osteophagia (chewing carcass bones)
TETANUS	<i>Cl. tetani</i> — enters through a deep puncture wound . Tetanospasmin blocks the release of the inhibitory transmitters glycine and GABA from the Renshaw cells → SPASTIC paralysis : lockjaw (trismus), " sawhorse " stance, erect ears, protrusion of the third eyelid , spasms triggered by noise or touch; the animal remains conscious . Horses are the most susceptible ; there are no diagnostic gross lesions

TWO TOXINS, OPPOSITE RESULTS **BOTULISM = FLACCID** (ACh release blocked). **TETANUS = SPASTIC/RIGID** (inhibition blocked). Both are ***Clostridium*** neurotoxins.

12 · ACTINOMYCOSIS · STREPTOCOCCI · STAPHYLOCOCCI · GLANDERS

- **ACTINOMYCOSIS** — "lumpy jaw": ***Actinomyces bovis***, a rarefying and proliferative **osteomyelitis of the MANDIBLE**; pyogranuloma containing "**sulphur granules**" (club colonies) with the **Splendore–Hoeppli** reaction
- **ACTINOBACILLOSIS** — "wooden tongue": ***Actinobacillus lignieresii***, in **SOFT tissue** — the tongue becomes hard and protrudes; same club colonies
- **STRANGLES**: ***Streptococcus equi* subsp. *equi*** in young horses — fever, nasal discharge and **abscessation of the submandibular and retropharyngeal lymph nodes** which rupture and discharge thick pus. "**Bastard strangles**" = metastatic abscesses in other organs. **PURPURA HAEMORRHAGICA** is the **Type III immune-complex sequel** — widespread petechiae and severe oedema
- **STAPHYLOCOCCOSIS**: ***Staph. aureus*** — abscess, **gangrenous mastitis**, tick pyaemia of lambs, **botryomycosis** (chronic granuloma with club colonies)
- **GLANDERS** ★: ***Burkholderia mallei***, of equines, **zoonotic and notifiable in India**. **Nasal form** — ulcers on the septum that heal into a typical **stellate (star-shaped) scar**. **Pulmonary form** — miliary grey nodules. **Cutaneous form = "FARCY"** — **corded, thickened lymphatics studded with nodules ("farcy buds") that ulcerate**. Diagnosed by the **MALLEIN test** and complement fixation

13 · PASTEURELLOSIS ★★★ (TOP INDIAN TOPIC)

	Haemorrhagic septicaemia	Shipping fever / pneumonic pasteurellosis
Agent	<i>Pasteurella multocida</i> serotypes B:2 and E:2	<i>Mannheimia (Pasteurella) haemolytica</i> A1
Species	Cattle and especially BUFFALO ; monsoon disease of India	Cattle, sheep after transport/stress
Course	Peracute septicaemia , death in 6–24 h	Acute bronchopneumonia
Lesions	Hot painful SUBCUTANEOUS OEDEMA of the throat, brisket and dewlap (the "septicaemic/oedematous" form); widespread petechiae on serosae ; blood-tinged fluid in the body cavities; enlarged haemorrhagic lymph nodes; fibrinous pneumonia in the longer cases	CRANIOVENTRAL fibrinous bronchopneumonia with FIBRINOUS PLEURITIS ; interlobular oedema; the lung is red, firm and heavy with fibrin sheets on the pleura
Diagnosis	Bipolar organisms on a Leishman-stained blood smear ; culture from heart blood, spleen, oedema fluid	Culture from lung; history of transport

FIELD RULE A buffalo that dies in **24 hours during the rains with a swollen throat = HS**. A buffalo found dead with **tarry unclotted blood and a huge soft spleen = ANTHRAX**. Do not confuse them — the second one you must not open.

14 · LEPTOSPIROSIS · LISTERIOSIS · SWINE ERYSIPELAS · BRUCELLOSIS

Disease	Agent & lesions
LEPTOSPIROSIS	<i>Leptospira interrogans</i> serovars — <i>pomona</i> , <i>hardjo</i> , <i>canicola</i> , <i>icterohaemorrhagiae</i> , <i>grippotyphosa</i> . Spread in URINE , enters through wet skin and mucosa; rodents are the reservoir . Acute in calves — intravascular haemolysis with haemoglobinuria ("REDWATER") , jaundice and fever. Chronic — interstitial nephritis giving the "WHITE-SPOTTED KIDNEY" , abortion, and agalactia with flabby udder in cows . In dogs, acute nephritis and hepatitis. Diagnosis: dark-field microscopy of urine , MAT , silver stain of kidney, PCR. Important ZOONOSIS (Weil's disease)
LISTERIOSIS	<i>Listeria monocytogenes</i> — Gram-positive, grows at refrigeration temperature (cold enrichment) and multiplies in poorly fermented SILAGE with pH above about 5.5 . Three syndromes: ① "CIRCLING DISEASE" — ascends the trigeminal nerve to give MICROABSCESSES IN THE BRAIN STEM (pons and medulla) with unilateral cranial nerve deficits, drooping ear and lip, circling. ② Abortion in the last trimester with focal hepatic necrosis in the fetus. ③ Septicaemia with miliary hepatic necrosis in monogastrics and neonates. Zoonotic
SWINE ERYSIPELAS	<i>Erysipelothrix rhusiopathiae</i> . Acute — septicaemia with DIAMOND-SHAPED (rhomboid) raised red-purple SKIN LESIONS , pathognomonic. Chronic — vegetative endocarditis of the LEFT AV (mitral) valve and proliferative non-suppurative arthritis . Zoonotic (erysipeloid in fish handlers)
BRUCELLOSIS ★	<i>Brucella abortus</i> (cattle), <i>B. melitensis</i> (goats, sheep — the most pathogenic for man), <i>B. suis</i> , <i>B. ovis</i> (rams). Facultative intracellular , survives in macrophages; has a predilection for the placenta and fetal fluids because of ERYTHRITOL . Lesions: abortion in the LAST TRIMESTER , retained placenta and metritis; necrotic yellow leathery cotyledons with a THICKENED, OEDEMATOUS INTERCOTYLEDONARY MEMBRANE covered in sticky exudate; fetal bronchopneumonia ; in bulls orchitis and epididymitis with testicular necrosis and seminal vesiculitis ; carpal/knee hygroma in Indian cattle. Tests: RBPT (screening) , STAT , milk ring test , CFT , ELISA ; culture of fetal abomasal content is the best sample. Zoonosis — undulant/Malta fever

15 · CORYNEBACTERIUM · CAMPYLOBACTER · SALMONELLA · COLI · NECROBACILLOSIS

Disease	Key point
Caseous lymphadenitis	<i>Corynebacterium pseudotuberculosis</i> , sheep and goats; abscess of superficial nodes and lung with concentric laminated "ONION-RING" pus
Bovine pyelonephritis	<i>Corynebacterium renale</i> — ascending infection; pus in the renal pelvis with radiating pale streaks in the medulla; also pizzle rot of wethers
CAMPYLOBACTERIOSIS	<i>Campylobacter fetus</i> subsp. <i>venerealis</i> — bovine genital campylobacteriosis ; venereal , causes early embryonic death , repeat breeding and infertility rather than late abortion; mild endometritis. <i>C. jejuni</i> — enteritis and ovine abortion ; a major food-borne zoonosis
SALMONELLOSIS	<i>S. Dublin</i> and <i>S. Typhimurium</i> in calves; <i>S. Abortusequi</i> (equine abortion), <i>S. Abortusovis</i> . Lesions: fibrinonecrotic ("diphtheritic") enterocolitis with a yellow-grey adherent pseudomembrane, "PARATYPHOID NODULES" = pinpoint pale foci of necrosis in the liver , cholecystitis with a thickened gall bladder, splenomegaly, septicaemia in the young. Zoonosis
COLIBACILLOSIS	<i>E. coli</i> . ① Enterotoxigenic (ETEC) — fimbriae K99 (F5) in calves and lambs , K88 (F4) and F18 in piglets ; enterotoxin causes SECRETORY diarrhoea with a NORMAL villus (no villous atrophy) → profuse watery white diarrhoea, dehydration, acidosis. ② Septicaemic colibacillosis — follows failure of passive transfer ; polyarthritis, meningitis, omphalitis. ③ OEDEMA DISEASE of pigs ★ — Shiga toxin 2e from F18 <i>E. coli</i> soon after weaning ; oedema of the EYELIDS , forehead , submucosa of the GASTRIC CARDIA and MESOCOLON , hydrothorax and ascites; staggering, squealing voice, sudden death; angiopathy with fibrinoid necrosis. ④ Coliform mastitis
NECROBACILLOSIS	<i>Fusobacterium necrophorum</i> — the common denominator of calf diphtheria (necrotic laryngitis and oral necrosis), foot rot (with <i>Dichelobacter nodosus</i>), rumenitis–liver abscess complex , and necrotic rhinitis (bull nose) of pigs . Foul odour and yellow-grey dry necrotic tissue

16 · MYCOPLASMA · CHLAMYDIA · Q-FEVER · ANAPLASMA · EHRlichia ★★

Disease	Agent & lesion
CBPP Contagious bovine pleuropneumonia	<i>Mycoplasma mycoides</i> subsp. <i>mycoides</i> (small colony type). Notifiable. Unilateral fibrinous pleuropneumonia: massive thickening of the INTERLOBULAR SEPTA with hepatisation of different ages gives the classic "MARBLED" cut surface; litres of straw-coloured fluid with fibrin clots in the pleural cavity; chronic cases form SEQUESTRA — encapsulated necrotic lung that keeps live organisms and makes carriers
CCPP Contagious caprine pleuropneumonia	<i>Mycoplasma capripneumoniae</i> in goats; acute fibrinous pleuropneumonia, usually unilateral, with pleural fluid; high mortality; no marbling and no sequestra as a rule
Porcine enzootic pneumonia	<i>Mycoplasma hyopneumoniae</i> . Cranioventral purple-red plum-coloured consolidation with sharp demarcation; the histological hallmark is "CUFFING PNEUMONIA" — heavy peribronchiolar and perivascular LYMPHOID hyperplasia. High morbidity, low mortality, poor growth
Chlamydial diseases	<i>Chlamydia abortus</i> → enzootic abortion of ewes, necrotic placentitis. <i>C. psittaci</i> → psittacosis/ornithosis in birds — fibrinous airsacculitis, pericarditis, hepatosplenomegaly; zoonotic. Also polyarthritis, conjunctivitis, sporadic bovine encephalomyelitis. Obligate intracellular; elementary and reticulate bodies, seen with Giemsa or Macchiavello stain
Q FEVER	<i>Coxiella burnetii</i> — extremely resistant, spread by aerosol from birth fluids and by ticks; causes abortion and placentitis in ruminants (often subclinical) but serious influenza-like disease and endocarditis in people. A major occupational zoonosis of veterinarians
ANAPLASMOSIS ★	<i>Anaplasma marginale</i> in the red cell, tick-borne. Causes EXTRAVASCULAR haemolysis — therefore marked anaemia and JAUNDICE but NO haemoglobinuria. Pale thin blood, enlarged spleen, distended gall bladder, icteric carcass. Seen as a small dense dot at the MARGIN of the RBC
EHRlichiosis	<i>Ehrlichia canis</i> → tropical canine pancytopenia, spread by <i>Rhipicephalus sanguineus</i> . Thrombocytopenia with epistaxis and petechiae, anaemia, lymphadenopathy, splenomegaly; MORULAE in monocytes. Common in Indian dogs

HAEMOGLOBINURIA — YES OR NO? ★★★ BABESIOSIS — YES (intravascular haemolysis, red urine). ANAPLASMOSIS — NO (extravascular haemolysis, jaundice only). THEILERIOSIS — usually NO. Leptospirosis and post-parturient haemoglobinuria and chronic copper poisoning — YES. This one line settles many objective questions.

17 · SUPERFICIAL & DEEP MYCOSES

Mycosis	Agent & diagnostic feature
RINGWORM (dermatophytosis)	<i>Trichophyton verrucosum</i> — the usual cause in CATTLE, thick grey-white crusty circular plaques around the eyes and neck. <i>Microsporum canis</i> — dogs and cats, and the one that shows apple-green FLUORESCENCE under WOOD'S (UV) LAMP. <i>T. mentagrophytes</i> , <i>M. gypseum</i> . Lesion: circular expanding alopecia with scaling and a healing centre; hyphae and arthrospores confined to the keratin and hair shaft, folliculitis. Diagnosis: 10–20% KOH mount of hair/scale, Wood's lamp, culture on DTM/Sabouraud. Zoonotic
ASPERGILLOSIS	<i>Aspergillus fumigatus</i> ; septate, dichotomously branching hyphae at about 45°. Bovine mycotic abortion ★ — thickened leathery necrotic placenta with raised plaques, and discrete raised "ringworm-like" plaques on the SKIN of the aborted fetus. Guttural pouch mycosis of horses — erodes the internal carotid → fatal epistaxis. Brooder pneumonia in chicks. Angioinvasive with infarction. Stain: PAS or GMS
ZYGOMYCOSIS (mucormycosis)	<i>Rhizopus</i> , <i>Mucor</i> , <i>Absidia</i> . Broad, irregular, NON-SEPTATE (sparsely septate) hyphae branching at wide angles. Strongly angioinvasive → thrombosis and infarction; gastric and abomasal ulcers, lymph node granulomas, mycotic placentitis
HISTOPLASMOSIS	<i>Histoplasma capsulatum</i> — small intracellular yeasts inside macrophages; granulomatous disease of the intestine, liver, spleen and lung of dogs. <i>H. farciminosum</i> causes EPIZOOTIC LYMPHANGITIS of horses — corded lymphatics with ulcerating nodules (must be told apart from glanders)
CRYPTOCOCCOSIS	<i>Cryptococcus neoformans</i> , associated with pigeon droppings. Yeast with a huge mucoid CAPSULE that produces a gelatinous, "soap-bubble" appearance. Nasal granuloma of the CAT ("Roman nose"), meningitis. Capsule shown by INDIA INK negative staining and by MUCICARMINE (positive) — a favourite question
RHINOSPORIDIOSIS ★	<i>Rhinosporidium seeberi</i> — very common in INDIA (cattle, buffalo, horses, dogs and man), linked to stagnant water and mud. Pink, polypoid, friable nasal mass studded with white pinhead specks, which are the SPORANGIA packed with endospores. Bleeds easily, recurs after removal
CANDIDIASIS	<i>Candida albicans</i> — "THRUSH", a white pseudomembrane on the tongue and oesophagus; crop mycosis (sour crop) in birds; follows prolonged antibiotic use and immunosuppression. Shows budding yeasts, pseudohyphae and hyphae together

18 · MYCOTOXICOSES ★★★

Mycotoxycosis	Fungus, toxin, target
AFLATOXICOSIS	<i>Aspergillus flavus</i> and <i>A. parasiticus</i> → aflatoxin B ₁ , the most toxic. On groundnut cake, maize and cottonseed stored damp — the classic Indian feed problem. Target = LIVER; it is both hepatotoxic and HEPATOCARCINOGENIC. Acute: centrilobular necrosis and haemorrhage, icterus, bleeding (↓ clotting factors). Chronic: firm pale fibrotic liver with marked BILE DUCT (biliary) HYPERPLASIA, MEGALOCYTOSIS of hepatocytes, and eventually hepatocellular carcinoma . Ducklings are the most sensitive species; also immunosuppression and poor growth. Aflatoxin M ₁ is excreted in milk
OCHRATOXICOSIS	<i>Aspergillus ochraceus</i> , <i>Penicillium verrucosum</i> → ochratoxin A . Target = KIDNEY (nephrotoxic). Pale, enlarged, mottled kidneys with tubular degeneration and interstitial fibrosis — "mycotoxic porcine nephropathy"; polyuria, poor growth. Immunosuppressive
TRICHOTHECOSIS	<i>Fusarium</i> species → T-2 toxin, deoxynivalenol (vomitoxin), diacetoxyscirpenol . Trichothecenes are protein-synthesis inhibitors and are RADIOMIMETIC: oral and gastrointestinal necrosis and ulceration , feed refusal and vomiting, plus necrosis of bone marrow and lymphoid tissue → pancytopenia and haemorrhage . Also contact dermatitis of the muzzle
DEGNALA DISEASE ★	An Indian condition of buffaloes and cattle in the northern states , linked to feeding mouldy PADDY (rice) STRAW and its <i>Fusarium</i> toxins. Produces DRY GANGRENE OF THE EXTREMITIES — oedema then necrosis and sloughing of the tail tip, ear tips, muzzle and coronary band/hoof , with sharply demarcated dry black tissue. Not febrile; response is reported to selenium plus vitamin E and change of fodder
ERGOTISM	<i>Claviceps purpurea</i> on rye and other grasses → ergot alkaloids (ergotamine, ergotamine) which are powerful VASOCONSTRICTORS → DRY GANGRENE of the ear tips, tail, teats and feet , agalactia, abortion; a convulsive/nervous form also occurs. <i>Claviceps fusiformis</i> on BAJRA (pearl millet) in India causes AGALACTIA in sows and buffaloes

ONE-LINE TARGETS Aflatoxin → **LIVER**. Ochratoxin → **KIDNEY**. Trichothecene → **GUT + BONE MARROW**. Fumonisin → **BRAIN** of the horse (leukoencephalomalacia). Zearalenone → **OESTROGENIC** (vulvar oedema, prolapse in gilts). Ergot → **BLOOD VESSELS** (gangrene).

19 · HAEMOPROTOZOA ★★★ (HIGHEST YIELD IN INDIA)

	BABESIOSIS	THEILERIOSIS	TRYPANOSOMOSIS
Agent	<i>Babesia bigemina</i> , <i>B. bovis</i> ; <i>B. canis</i> in dogs	<i>Theileria annulata</i> (tropical theileriosis of India), <i>T. parva</i> (East Coast fever)	<i>Trypanosoma evansi</i> = SURRA ; also <i>T. congolense</i> , <i>T. vivax</i>
Vector	<i>Rhipicephalus (Boophilus) microplus</i> tick; transovarial	<i>Hyalomma</i> tick; transstadial	Biting flies (Tabanus) MECHANICALLY ; also iatrogenic needles
Where it lives	Inside the RBC — large PEAR-SHAPED PAIRS at an acute angle	Schizonts (KOCH'S BLUE BODIES) in LYMPHOCYTES , then small piroplasms in RBCs	EXTRACELLULAR flagellate in plasma
Key signs	High fever, intravascular haemolysis → HAEMOGLOBINURIA ("redwater") , jaundice, anaemia. <i>B. bovis</i> also causes CEREBRAL babesiosis — parasitised cells sequester in brain capillaries, giving a "cherry-pink" brain and nervous signs	Marked enlargement of superficial lymph nodes first (parotid node), fever, anaemia, jaundice, petechiae, "PUNCHED-OUT" ABOMASAL ULCERS , pulmonary oedema, lymphoid proliferation then destruction . Exotic breeds and crossbreds worst	Intermittent fever, progressive anaemia and cachexia , ventral oedema, corneal opacity in dogs , splenomegaly; nervous signs late. Horses, camels and dogs are highly susceptible ; cattle and buffalo are often carriers
Diagnosis	Giemsa-stained peripheral blood smear from the EAR-TIP capillary	Lymph node biopsy/aspirate smear for Koch's blue bodies + blood smear	Wet blood film for motile organisms , Giemsa smear, mouse inoculation , CATT, PCR

20 · FLUKES, WORMS, TAPEWORMS, COCCIDIA

Parasite	Pathology
FASCIOLOSIS ★	<i>Fasciola hepatica</i> and <i>F. gigantica</i> (the important one in India); snail host <i>Lymnaea</i> . Acute: immature flukes tunnel through the parenchyma → haemorrhagic migratory tracts , enlarged friable liver, fatal haemorrhage; also creates the anaerobic focus for BLACK DISEASE . Chronic: adults in the bile ducts → cholangitis, marked biliary fibrosis and calcification = " PIPE-STEM LIVER ", cirrhosis, black parasitic haematin pigment, anaemia, hypoalbuminaemia and BOTTLE JAW
AMPHISTOMIASIS	<i>Paramphistomum</i> . The IMMATURE flukes are the killers — they attach in the duodenum causing severe haemorrhagic duodenitis , oedema and death in young stock; the ADULTS in the rumen are relatively harmless (rumen papillae studded with pink flukes)
Ascariasis	<i>Ascaris suum</i> in pigs — larval migration leaves " MILK SPOTS " (white fibrotic foci) on the LIVER , a condemnation lesion; larvae also cause verminous pneumonia. <i>Parascaris equorum</i> in foals — intestinal obstruction and rupture. <i>Toxocara canis</i> — zoonotic visceral larva migrans
Strongylosis	<i>Strongylus vulgaris</i> — larvae migrate in the CRANIAL MESENTERIC ARTERY → verminous arteritis and aneurysm , thrombosis, intestinal infarction and fatal colic in horses
HAEMONCHOSIS	<i>Haemonchus contortus</i> , the " BARBER'S POLE WORM " of the abomasum . A voracious blood-sucker → severe anaemia, hypoproteinaemia and BOTTLE JAW , usually WITHOUT diarrhoea ; pale watery carcass. Abomasal mucosa studded with worms and small bleeding points
SPIROCERCOSIS	<i>Spirocerca lupi</i> in the dog's OESOPHAGUS (beetle intermediate host) → firm nodules with a central pore that may transform into FIBROSARCOMA or OSTEOSARCOMA ; also aortic scarring/aneurysm and spondylitis of the caudal thoracic vertebrae
Filariasis	<i>Setaria</i> — harmless in the peritoneum but aberrant migration into the spinal cord causes CEREBROSPINAL NEMATODIASIS ("KUMRI") in goats and horses in India — sudden hind-limb weakness. <i>Dirofilaria immitis</i> — heartworm in the right ventricle and pulmonary artery , pulmonary endarteritis, caval syndrome . <i>Parafilaria bovicola</i> — "summer bleeding" nodules on cattle skin. <i>Onchocerca</i> — ligament and skin
Hookworm	<i>Ancylostoma caninum</i> — severe iron-deficiency anaemia and melena in pups; percutaneous, transmammary and oral routes
Tapeworms	<i>Cysticercus bovis</i> (<i>Taenia saginata</i>) in beef muscle, heart and tongue — "measly beef". <i>Cysticercus cellulosae</i> (<i>T. solium</i>) in pork — and human neurocysticercosis . <i>Echinococcus granulosus</i> → HYDATID CYST in the liver and lung of herbivores and man; the dog is the definitive host ; the cyst has a laminated membrane and brood capsules with hydatid sand . <i>Coenurus cerebralis</i> (<i>T. multiceps</i>) → " GID "/sturdy in sheep brain. <i>Moniezia</i> in ruminants, usually mild
Coccidiosis	<i>Eimeria zuernii</i> and <i>E. bovis</i> in calves — haemorrhagic typhlocolitis , bloody diarrhoea with tenesmus and rectal prolapse ; the parasite destroys the epithelium of the caecum and colon
Toxoplasmosis	<i>Toxoplasma gondii</i> ; the CAT is the only definitive host (sheds oocysts). Causes abortion in SHEEP with small WHITE NECROTIC FOCI scattered over the COTYLEDONS ("strawberry jam" placenta) , fetal brain and liver necrosis; encephalitis and pneumonia in other species. Zoonosis — dangerous in pregnancy
Cryptosporidiosis	<i>Cryptosporidium parvum</i> — a major cause of calf diarrhoea at 1–3 weeks ; attaches to the microvillous border of the ileum → villous atrophy and malabsorption . Oocysts are acid-fast (modified Ziehl-Neelsen) ; water-borne ZOONOSIS, resistant to chlorine

21 · NUTRITIONAL IMBALANCES ★★

Energy & macronutrients — Carbohydrate excess → **ruminal lactic acidosis** (grain overload) with rumenitis and liver abscess; carbohydrate deficit → **ketosis** and pregnancy toxemia. **Protein deficiency** → hypoproteinaemic **oedema and ascites**, poor growth, immunosuppression. **Fat** — essential fatty acid deficiency gives scaly skin; **steatitis ("yellow fat disease")** occurs in cats fed oily fish with **vitamin E deficiency**.

Mineral	Deficiency disease
Ca / P	Rickets (young), osteomalacia (adult), fibrous osteodystrophy when Ca:P is wrong, milk fever (parturient hypocalcaemia)
Magnesium	HYPOMAGNESAEMIC TETANY ("grass tetany/staggers") — lush spring grass, lactating cows; tetany, convulsions, sudden death
COPPER	SWAY BACK / enzootic ataxia in lambs (cerebral white-matter cavitation, spinal demyelination) · achromotrichia — loss of coat colour with "SPECTACLE EYE" in cattle · anaemia · "falling disease" from myocardial fibrosis · poor wool crimp ("steely wool"). Made worse by excess molybdenum and sulphate
Cobalt	Needed for vitamin B ₁₂ synthesis in the rumen → "pine"/wasting , and WHITE LIVER DISEASE of lambs (fatty liver)
Iodine	GOITRE — enlarged thyroid; newborns weak, HAIRLESS, stillborn with a swollen neck ; goitrogens in brassicas and soybean
SELENIUM (+ vit E)	WHITE MUSCLE DISEASE, hepatosis dietetica of pigs, MULBERRY HEART DISEASE of pigs, exudative diathesis of chicks, retained placenta, poor immunity
Zinc	PARAKERATOSIS of pigs — symmetrical crusty scaly skin; thymic atrophy, poor wound healing
Iron	Baby-pig anaemia — microcytic hypochromic; the classic reason for iron injection at birth
Vitamin	Deficiency lesion
A	Night blindness → xerophthalmia, SQUAMOUS METAPLASIA and keratinisation of epithelium, hydrocephalus and blindness from stenosis of the optic canal in calves , pituitary cysts, poor immunity, papilloedema
D	Rickets and osteomalacia
E	See selenium; also steatitis and reproductive failure
K	Bleeding — sweet clover (dicoumarol), warfarin, prolonged sulphonamide use
B₁ thiamine	POLIOENCEPHALOMALACIA in ruminants (thiaminase, high grain, amprolium, bracken fern in horses); "Chastek paralysis" in fur animals and cats fed raw fish thiaminase
B₂ riboflavin	"CURLED-TOE PARALYSIS" of chicks — peripheral nerve demyelination
Niacin	"BLACK TONGUE" of dogs, pellagra-like; maize-based diets
Choline / Manganese	PEROSIS ("slipped tendon") in chicks
C	SCURVY — only in GUINEA PIGS, primates and fish , which cannot synthesise it; haemorrhage, poor collagen, loose teeth

22 · METABOLIC DISEASES ★★

Disease	Mechanism & lesion
PREGNANCY TOXAEMIA	Ewes and does in LATE gestation carrying twins/triplets ; negative energy balance → fat mobilisation → ketosis and severe FATTY LIVER (huge, pale yellow, greasy, friable). Ketonuria, hypoglycaemia, blindness, recumbency. Also seen in fat queens
Ketosis of dairy cattle	Early lactation, high yield; ketone smell on breath and milk, fatty liver, low milk yield; Rothera's test on urine/milk
POST-PARTURIENT HAEMOGLOBINURIA ★	HYPOPHOSPHATAEMIA in high-yielding buffaloes and cows within a few weeks of calving , often on rice straw/berseem with low phosphorus. Intravascular haemolysis → haemoglobinuria, severe anaemia, pale mucosae, jaundice, pale fatty liver . Responds to phosphorus therapy. Distinguish from babesiosis (which has parasites on the smear and fever)
Hypomagnesaemic tetany	As above; often no gross lesion — diagnosis from serum/CSF magnesium and history
AZOTURIA	Exertional rhabdomyolysis of horses: full grain ration during rest, then sudden work. Hard, painful, swollen gluteal muscles , sweating, reluctance to move, coffee-coloured MYOGLOBINURIC urine ; pale streaked muscle at necropsy; ↑ CK and AST. Plasma remains clear
SWAY BACK / enzootic ataxia	Copper deficiency in lambs — congenital form present at birth, delayed form later; white-matter cavitation and demyelination
PICA	Depraved appetite from phosphorus and mineral deficiency ; chewing bones and soil → leads to BOTULISM and foreign-body/hardware disease
Rheumatism-like syndrome	Stiffness, shifting lameness and joint pain without a single defined agent; polyarthritis, periarticular fibrosis

24 • PLANT POISONING ★★★

Plant	Toxin & pathology
BRACKEN FERN <i>Pteridium aquilinum</i>	Toxin PTAQUILOSIDE (radiomimetic) plus THIAMINASE . Three quite different pictures — know all three: CATTLE (acute) — bone marrow APLASIA → pancytopenia, thrombocytopenia and a haemorrhagic syndrome with bleeding from all orifices and secondary infection. CATTLE (chronic) → ENZOOTIC HAEMATURIA — bladder haemangiomas and carcinomas with persistent haematuria (bovine papillomavirus acts as a co-factor). HORSES — thiaminase → thiamine deficiency , ataxia, "bracken staggers". SHEEP — "BRIGHT BLINDNESS" from progressive retinal degeneration
GOSSYPOL (cottonseed cake)	A polyphenol in cottonseed ; toxic to monogastrics and young ruminants (adult rumen binds it). Cardiotoxic and hepatotoxic → myocardial degeneration and heart failure, giving generalised oedema: pulmonary oedema, hydrothorax, hydropericardium and ascites ; centrilobular hepatic necrosis; anaemia; reduced fertility in males ; in layers, olive-green/discoLOURED yolks and mottled eggs
RATTI <i>Abrus precatorius</i>	The red-and-black "crab's-eye" seed. Toxin ABRIN , a toxalbumin / ribosome-inactivating protein (like ricin). In India the crushed seed paste is made into a "SUI" (needle) and driven under the skin to poison cattle deliberately — a veterolegal problem. Produces local oedema and necrosis at the site , then severe haemorrhagic gastroenteritis, agglutination and haemolysis of red cells, hepatic and renal degeneration and death in 1–3 days. At necropsy search the skin for the puncture and recover the needle
LANTANA <i>Lantana camara</i>	Toxin LANTADENE A (and B) — the classic Indian hepatotoxin of cattle, buffalo and sheep browsing the weed. Causes intrahepatic CHOLESTASIS and hepatocellular necrosis → jaundice, ruminal stasis and severe constipation , and above all HEPATOGENOUS (Type III) PHOTSENSITIZATION because phylloerythrin is no longer excreted: crusting, peeling and necrosis of the muzzle ("PINK NOSE") , eyelids, ears and any unpigmented skin . Also renal tubular damage. Liver is enlarged, yellow-orange and firm; gall bladder distended

HYDROCYANIC ACID ★	CHOCOLATE-BROWN , mucosae are muddy-brown/cyanotic, dyspnoea, rapid death; also abortion. Antidote: METHYLENE BLUE (and ascorbic acid)
FLUORIDE	Cyanogenic glycosides in young, wilted, drought-stressed or frosted SORGHUM/Johnson grass and cassava . Cyanide blocks cytochrome oxidase , so tissues cannot use oxygen. Blood stays BRIGHT CHERRY-RED , mucosae bright red, bitter-almond odour of the rumen contents, death in minutes. Antidote: sodium nitrite followed by sodium thiosulphate
SELENIUM (excess)	Dental fluorosis — mottled, chalky, pitted, brown-stained, rapidly worn teeth, but ONLY those teeth that were developing during exposure ; exostoses (bony outgrowths) on the mandible and metatarsals , stiffness and lameness. Sources: industrial fallout and high-fluoride bore-well water — endemic in parts of India
OXALATES	"ALKALI DISEASE" (chronic) — loss of hair from the mane and tail , and deformed, sloughing hooves with a circular break at the coronet . "BLIND STAGGERS" (subacute) — wandering, blindness, paralysis. From selenium-accumulating plants on seleniferous soil (Punjab/Haryana belt)
ORGANOPHOSPHATE / CARBAMATE	In Oxalis, Setaria, Halogeton, amaranthus, sugar beet tops . Bind calcium → acute hypocalcaemia and tetany , and calcium oxalate crystals in the renal tubules → NEPHROSIS and renal failure . Chronic low intake with grass leads to bone changes

ORGANOCHLORINE

oedema; nicotinic — muscle tremor then paralysis; CNS — convulsions. Diagnosis: **low blood cholinesterase**. **Antidote: ATROPINE** add **2-PAM (pralidoxime)** for **organophosphates but NOT for carbamates**

DDT, BHC/lindane, endosulfan — **CNS STIMULATION**: hyperexcitability, muscle tremors beginning at the head, exaggerated response to stimuli, convulsions. Highly **lipophilic and persistent** — **residues in fat and milk**. No specific antidote

25 • BLANK-FILL DRILL — UNIT 4

- FMD virus is an **Aphthovirus** of family **Picornaviridae** with **7** serotypes
- Serotypes of FMD circulating in India are **O, A and Asia-1**
- Tiger heart in calves occurs in **FMD**
- Of the vesicular diseases, only **vesicular stomatitis** affects horses
- Zebra striping of the colon is seen in **rinderpest** and **PPR**
- The microscopic hallmark of malignant catarrhal fever is **necrotising lymphocytic vasculitis**
- Bluetongue is spread by **Culicoides** and shows haemorrhage at the base of the **pulmonary artery**
- IBR virus becomes latent in the **trigeminal ganglion**
- Persistently infected BVD calves arise from **non-cytopathic** infection at **40–120** days of gestation
- Linear oesophageal erosions suggest **mucosal disease**
- EIA is diagnosed by the **Coggins** test and shows **sideroleukocytes**
- EHV-1 abortion produces a **fresh** (non-autolysed) fetus with hepatic **intranuclear** inclusions
- Button ulcers, marginal splenic infarcts and turkey-egg kidney = **classical swine fever**
- Rabies virus travels to the CNS **centripetally** along **peripheral nerves**
- The gold-standard test for rabies is the **fluorescent antibody test** on brain
- Hard pad and enamel hypoplasia occur in **canine distemper**
- Gall-bladder wall oedema with Cowdry type A inclusions = **infectious canine hepatitis**
- Canine parvovirus destroys the intestinal **crypts**; rotavirus destroys the **villus tips**
- The wheelbarrow test is used for **jaagsiekte** (ovine pulmonary adenocarcinoma)
- Prions consist of **protein** only and cause **status spongiosus** with **no** inflammation
- Bollinger bodies occur in **fowl pox**; Guarnieri bodies in **vaccinia**
- Orf is caused by a **parapoxvirus** and is **zoonotic**
- Pearl disease refers to serosal **tuberculosis** in cattle
- Johne's disease shows **no** caseation and **no** ulceration, unlike tuberculosis
- Blackleg is **endogenous** and needs no wound; malignant oedema is **exogenous**
- Black disease requires liver damage by immature **Fasciola**
- Pulpy kidney of lambs is caused by **Clostridium perfringens type D** and its **epsilon** toxin
- Botulism causes **flaccid** paralysis; tetanus causes **spastic** paralysis
- Tetanospasmin blocks the release of **glycine and GABA**
- Star-shaped nasal scars and farcy buds occur in **glanders**; test = **mallein**
- Haemorrhagic septicaemia is caused by **Pasteurella multocida** serotypes **B:2 and E:2**
- Shipping fever is caused by **Mannheimia haemolytica**
- Diamond-shaped skin lesions = **swine erysipelas**
- Brucella localises in the placenta because of **erythritol**; screening test = **RBPT**
- Circling disease from brain-stem microabscesses = **listeriosis**, from feeding **silage**
- Paratyphoid nodules in the liver indicate **salmonellosis**
- Oedema disease of pigs is caused by **Shiga toxin 2e** and affects the **eyelids and gastric cardia**
- Marbled lung with sequestra = **CBPP**; cuffing pneumonia = **Mycoplasma hyopneumoniae**
- Anaplasmosis gives anaemia and jaundice but **no** haemoglobinuria
- Morulae in monocytes indicate **Ehrlichia canis**
- Microsporium canis fluoresces under the **Wood's lamp**
- Aspergillus hyphae are **septate**; zygomycete hyphae are **non-septate**
- Cryptococcus capsule is shown by **India ink** and stains with **mucicarmine**
- Nasal polyps with sporangia = **rhinosporidiosis**
- Aflatoxin targets the **liver**; ochratoxin the **kidney**; trichothecenes the **gut and bone marrow**
- Chronic aflatoxicosis causes **bile duct hyperplasia** and **megalocytosis**
- Degenerative disease follows feeding mouldy **paddy straw** and causes **dry gangrene of the extremities**
- Ergot alkaloids cause gangrene by **vasoconstriction**; Claviceps fusiformis on **bajra** causes agalactia
- Babesia appears as **pear-shaped pairs**; Theileria schizonts are **Koch's blue bodies**

- Theileria annulata is transmitted by **Hyalomma** ticks and causes **punched-out abomasal ulcers**
- Surra is caused by **Trypanosoma evansi** spread **mechanically** by Tabanus flies
- Pipe-stem liver is chronic **fasciolosis**; the snail host is **Lymnaea**
- In amphistomiasis the **immature** flukes in the **duodenum** cause the fatal disease
- Milk spots on the pig liver are caused by **Ascaris suum** larvae
- Haemonchus contortus is the **barber's pole** worm of the **abomasum**
- Spirocerca lupi nodules may become **fibrosarcoma** or **osteosarcoma**
- Kumri in goats and horses is caused by aberrant **Setaria** larvae
- The definitive host of Echinococcus granulosus is the **dog**; the cyst contains **hydatid sand**
- Coenurus cerebralis causes **gid** in sheep
- Toxoplasma abortion in sheep shows white necrotic foci on the **cotyledons**; definitive host = **cat**
- Cryptosporidium oocysts are **acid-fast** and resistant to **chlorine**
- Curled-toe paralysis in chicks = **riboflavin** deficiency; perosis = **manganese/choline**
- Black tongue in dogs = **niacin** deficiency
- Scurvy occurs only in **guinea pigs, primates and fish**
- Spectacle eye in cattle indicates **copper** deficiency
- White liver disease of lambs = **cobalt** deficiency
- Post-parturient haemoglobinuria is caused by **hypophosphataemia**
- Rice-water diarrhoea with a red velvety abomasum = **arsenic** poisoning
- Gun-metal kidneys and a yellow friable liver in sheep = chronic **copper** poisoning
- Basophilic stippling with nucleated RBCs in a blind, head-pressing calf = **lead** poisoning
- Strychnine blocks **glycine**; convulsions are triggered by **noise, light or touch**
- Nitrite poisoning gives **chocolate-brown** blood; antidote **methylene blue**
- Cyanide poisoning gives **bright cherry-red** blood; antidote **sodium nitrite + sodium thiosulphate**
- Alkali disease is chronic **selenium** poisoning with loss of **mane and tail** hair
- Organophosphates inhibit **acetylcholinesterase**; antidote **atropine** plus **2-PAM**
- Bracken fern toxin is **ptaquiloside**; it causes **bright blindness** in sheep
- Gossypol comes from **cottonseed** and is **cardiotoxic**
- Ratti seed toxin is **abrin**, used in the veterolegal **sui** (needle) method
- Lantana toxin is **lantadene A** and causes **hepatogenous photosensitization**

